Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *CRISISLTLSUM* | ***Context:*** *Mexico City and Central America Bilingual Crisis Journalism*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's input ontology is restricted to four crisis domains — wildfire, local fire, traffic, and storm [Q1, Q53, Q122] — explicitly chosen because they are the most frequent in the selected US and Australian states [Q124]. For the target deployment, the elicitation and regional research identify earthquakes, volcanic activity, tropical-system flooding, civil unrest, and CDMX-specific transportation infrastructure failures as structurally important crisis types. Earthquakes are central to CDMX coverage given the 1985, 2017, and 2022 events, and Popocatépetl produces near-continuous low-level activity that yields a constant tweet stream rather than discrete spike events [WEB-7, WEB-20, WEB-26]. Even nominally overlapping categories (storms) carry different regional profiles: Mexican/Central American tropical systems use distinct alert vocabularies (semáforo phases, SMN/CONRED naming) that are not represented [WEB-19]. The 'local' definition (building/street/county, short-lived) [Q12] also conflicts with persistent volcanic phenomena and multi-day tropical systems. The taxonomy is therefore both under-representative (missing required categories) and structurally mismatched on the categories it does include.

ID	Evidence
Q1	“CrisisLTLSum contains 1,000 crisis event timelines across four domains: wildfires, local fires, traffic, and storms.” (p.1)
Q53	“The selected clusters cover events mainly from four crisis domains, including wildfire, local fire, storm, and traffic.” (p.5)
Q122	“We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm.” (p.12)
Q124	“These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York.” (p.13)
Q12	“Moreover, we focus on the extraction of local crisis events. The term "local" indicates that an event is bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period.” (p.2)
WEB-7	SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories (https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System)
WEB-20	CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises (https://www.cenapred.unam.mx/reportesVolcanesMX/)
WEB-26	Popocatépetl is at Amarillo Fase 2 with ongoing daily exhalations (167 on April 28, 2026), indicating persistent rather than spike-event tweet streams (https://www.infobae.com/mexico/2026/04/29/popocatepetl-tuvo-167-emisiones-este-28-de-abril/)
WEB-19	https://www.frontiersin.org/journals/earth-science/articles/10.3389/feart.2022.827236/full

Strengths

- Three of the four covered domains (local fire, traffic, storms) have at least nominal applicability to CDMX urban crisis coverage [Q53, Q122], so timeline/summarization patterns learned on those event types may partially transfer for those specific event types in urban contexts.

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Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the target region include earthquakes, volcanic activity (with semáforo phase vocabulary), tropical-system flooding, civil unrest/protests, organized-crime-related public-safety events, and CDMX transportation infrastructure failures, in addition to fire/traffic/storm [WEB-7, WEB-20, elicitation A1].

IO-2: Yes — earthquakes, volcanic activity, tropical-system flooding, and civil unrest are entirely absent from the benchmark's four-domain ontology [Q1, Q53, Q122]. The Popocatépetl semáforo vocabulary and SASMEX seed-tweet conventions have no analog in the benchmark [WEB-7, WEB-20].

IO-3: No category in the benchmark is wholly irrelevant to the target region — wildfires, local fires, traffic, and storms all have regional analogs [Q122] — but the 'storm' category as operationalized via US keyword lists [Q126, Q127] carries US-specific damage profiles that may not transfer.

IO-4: Construct underrepresentation is severe: four of the highest-priority crisis types in the deployment area (earthquake, volcanic, tropical flooding, civil unrest) are missing. The 'local' scope definition [Q12] also excludes persistent volcanic activity, which is structurally central to CDMX coverage [WEB-26].

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Q124	“These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York.” (p.13)
Q126	“We carefully curate a keyword list for each of our categories of interest by employing expert knowledge gathered through over-viewing crisis stories in news, social media, and related big disaster events of the past.” (p.13)
Q127	“Please note that these lists are generic for each category and not specific to unique events. For instance, instead of defining keywords specific to an event called "Hurricane Ida", our keywords list includes phrases such as "fallen tree", "building collapse", or "storm".” (p.13)
Q12	“Moreover, we focus on the extraction of local crisis events. The term "local" indicates that an event is bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period.” (p.2)
WEB-26	Popocatépetl is at Amarillo Fase 2 with ongoing daily exhalations (167 on April 28, 2026), indicating persistent rather than spike-event tweet streams (https://www.infobae.com/mexico/2026/04/29/popocatepetl-tuvo-167-emisiones-este-28-de-abril/)

Information Gaps

- Frequency of cartel/organized-crime crisis events in tweet streams is undocumented in indexed sources
- Relative weighting of missing categories vs. the four covered ones in actual journalist workflow not quantified

Requires Expert Verification

- Whether outlet-specific editorial priorities further alter category importance ranking

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Remediation

Gap 1: Earthquake, volcanic activity, tropical-system flooding, civil unrest, and CDMX transportation infrastructure failures are absent from the four-domain taxonomy [Q122]

Recommendation: Extend the ontology with these crisis types and curate Spanish-language seed-tweet/keyword lists for each, including SASMEX (#TenemosSismo, #AlertaSismica) [WEB-7] and CENAPRED semáforo vocabulary [WEB-20]. Recognize persistent (e.g., volcanic) vs. spike-event temporal profiles [WEB-26] in the 'local' definition [Q12].

Gap 2: The 'local' scope definition (building/street/county, short period) [Q12] excludes persistent volcanic activity and multi-day tropical systems

Recommendation: Generalize the temporal-scope definition to support persistent and multi-phase events, with explicit handling of semáforo phase transitions and tropical-system alert lifecycles.

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Q122	“We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm.” (p.12)
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